

**BAIDOO BONSOE SENIOR HIGH TECHNICAL SCHOOL**  
**FORM TWO, SEMESTER ONE (1)**  
**FACILITATOR NAME: BAAH EZEKIEL**

| Learning Plan                            |                                                                                                                                                                                                                                                                                                                                                  |                   |                                                                                           |                 |                    |             |          |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------|--------------------|-------------|----------|
| <b>Subject</b>                           | <i>Information &amp; Communications Technology</i>                                                                                                                                                                                                                                                                                               | <b>Week</b>       | <i>1</i>                                                                                  | <b>Duration</b> | <i>120 minutes</i> | <b>Form</b> | <i>2</i> |
| <b>Week Ending</b>                       | 9 <sup>th</sup> January, 2026                                                                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |
| <b>Strand</b>                            | <i>ICTs in the Society</i>                                                                                                                                                                                                                                                                                                                       | <b>Sub-Strand</b> | <i>Organizing, Managing and Presenting Information Using Essential Productivity Tools</i> |                 |                    |             |          |
| <b>Content Standard</b>                  | <i>Demonstrate knowledge and understanding of analyzing numerical data using Spreadsheet software.</i>                                                                                                                                                                                                                                           |                   |                                                                                           |                 |                    |             |          |
| <b>Learning Outcome(s)</b>               | <i>Utilize knowledge of Spreadsheet software to create functions for data analysis, interpret graphs and manipulate tables.</i>                                                                                                                                                                                                                  |                   |                                                                                           |                 |                    |             |          |
| <b>Learning Indicator(s)</b>             | <i>Describe Spreadsheet software workbooks, worksheets, cell referencing; save and print workbooks/worksheets.</i>                                                                                                                                                                                                                               |                   |                                                                                           |                 |                    |             |          |
| <b>Essential Question(s)</b>             | <ul style="list-style-type: none"> <li>▪ <i>What are the key features of spreadsheet software?</i></li> <li>▪ <i>How do workbooks, worksheets, and cell referencing help organize data?</i></li> <li>▪ <i>How can GESI, SEL, and national values (e.g., responsibility, accuracy) be integrated when teaching spreadsheet basics?</i></li> </ul> |                   |                                                                                           |                 |                    |             |          |
| <b>Pedagogical Strategies</b>            | <i>Talk for Learning</i><br><i>Experiential Learning</i><br><i>Brainstorming</i><br><i>Demonstration</i><br><i>Video-Assisted Learning</i><br><i>Group Work/Collaborative Learning</i><br><i>Think-Pair-Share, Questioning.</i>                                                                                                                  |                   |                                                                                           |                 |                    |             |          |
| <b>Teaching &amp; Learning Resources</b> | <i>Laptops/Desktop computers</i><br><i>MS Excel/Google Sheets</i><br><i>Projector, YouTube videos/OER (Excel interface tutorials)</i><br><i>USB drive</i><br><i>Printed handouts/steps</i>                                                                                                                                                       |                   |                                                                                           |                 |                    |             |          |
| <b>Key Notes on Differentiation</b>      |                                                                                                                                                                                                                                                                                                                                                  |                   |                                                                                           |                 |                    |             |          |

**i. Learning Task:**

Group learners considering GESI, SEL, and cross-cutting issues. Start with brainstorming real-life data organization (e.g., marks, budgets). Use mixed-ability/mixed-gender groups. Provide printed steps for approaching proficiency learners.

**ii. Pedagogical Exemplars:**

Think-Pair-Share (mixed groups discuss interface); Direct Instruction (demo interface); Video-Assisted Learning (Excel launch); Group practical (create/save workbook).

**iii. Key Assessments (DoK):****Level 1:**

Name spreadsheet parts.

**Level 2:**

Explain workbook vs. worksheet.

**Level 3:**

Describe steps to save/print.

**Level 4:**

Create, save, open, print a workbook unaided.

**Keywords**

***Spreadsheet, Workbook, Worksheet, Cell, Cell Reference, Row, Column, Formula Bar, Save, Save As, Print Preview.***

**Lesson 1****Focal Areas:**

1. Spreadsheet interface
2. Workbooks, worksheets, cell referencing
3. Basic file handling (create/save/open/print).

**Main Lesson drawing on Concepts, Skills and Competencies to reinforce as in the Subject Teacher Manual**

***Teacher Activity******Learner Activity*****Starter Activity (10 minutes)*****Brainstorming:***

*How do we organize numbers/data in daily life (e.g., school marks, market prices)?*

**Introductory Activity (15minutes)**

- Discuss spreadsheet uses
- Project/demo Excel interface (ribbons, cells, formula bar, workbook vs worksheet).

**Activity 1 (15 minutes)*****Demonstrate:***

*How to create workbook in excel.*

**Introductory Activity (15minutes)**

Observe demonstration and take note of key parts in notebooks.

**Activity 1**

*Practice creating new workbook; enter labels.*

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| <ul style="list-style-type: none"> <li>▪ <i>File &gt; New Blank workbook</i></li> <li>▪ <i>Rename worksheets</i></li> <li>▪ <i>Cell navigation/referencing (A1, B2).</i></li> </ul> <p><b><u>Activity 2 (15 minutes)</u></b><br/> <i>Video demo on file handling (Save/Save As, Open, Print).</i></p>                                                                                                        | <p><b><u>Activity 2</u></b><br/> <i>Watch, pause/replay, replicate steps.</i></p>                                                                                                                                                                                                                                               |
| <b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b>                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                 |
| <p style="text-align: center;"><b>Lesson Closure</b></p> <p style="text-align: center;"><i><b>In completing this part, refer to the Essential Questions to check that learning has taken place.</b></i></p>                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                 |
| <p><b><u>Activity (5 minutes)</u></b></p> <p><i>Recap with essential questions: "How does cell referencing support responsible data management?"</i></p>                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                 |
| <b>Lesson 2</b>                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                 |
| <p style="text-align: center;"><b>Focal Areas:</b></p> <p style="text-align: center;"><b>Basic data entry and cell referencing practice.</b></p>                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                 |
| <b><i>Teacher Activity</i></b>                                                                                                                                                                                                                                                                                                                                                                               | <b><i>Learner Activity</i></b>                                                                                                                                                                                                                                                                                                  |
| <p><b>Starter <u>Activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>• <i>Let students open saved workbook.</i></li> <li>• <i>Quick recap on the excel interface parts</i></li> </ul>                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                 |
| <p><b><u>Introductory activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>• <i>Demo data types (text, numbers, date and time).</i></li> <li>• <i>A walkthrough on Excel navigation shortcuts.</i></li> </ul> <p><b><u>Activity 1 (15 minutes)</u></b></p> <p><b><i>Guide practical:</i></b><br/> <i>Enter 10 cells with references</i></p> <p><b><u>Activity 2 (15 minutes)</u></b></p> | <p><b><u>Introductory activity</u></b><br/> <i>Enter names of teachers and students and few scores of students.</i></p> <p><b><u>Activity 1</u></b><br/> <i>Students will follow steps and save regularly.</i></p> <p><b><u>Activity 2 (15 minutes)</u></b><br/> <i>Students will discuss and add data collaboratively.</i></p> |

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| <p><i>Group discussion on real-life uses of excel and in-depth discussion gearing towards advance excel.</i></p>                                                                                                                                                                                                                                |  |
| <p><b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b></p>                                                                                                                                                                                                                                                               |  |
| <p><b>Level 4:</b></p> <ul style="list-style-type: none"> <li>• <i>Enter/save 20+ cells unaided.</i></li> <li>• <i>Explain one cell reference example</i></li> </ul>                                                                                                                                                                            |  |
| <p style="text-align: center;"><b>Lesson Closure</b></p> <p style="text-align: center;"><i><b>In completing this part, refer to the Essential Questions to check that learning has taken place.</b></i></p>                                                                                                                                     |  |
| <p><b><u>Activity (10 minutes)</u></b></p> <p><b>Recap:</b><br/><i>How does accurate referencing support ethical data use?</i></p>                                                                                                                                                                                                              |  |
| <p style="text-align: center;"><b>Reflection &amp; Remarks</b></p>                                                                                                                                                                                                                                                                              |  |
| <p><b>Reflection:</b></p> <ul style="list-style-type: none"> <li>• <i>High engagement in practical using mixed groups support GESI/SEL.</i></li> <li>• <i>There were limited laptops.</i></li> <li>• <i>Improvement can be made by printing more TLMS.</i></li> </ul> <p><b>Remarks:</b><br/><i>Lesson objectives are largely achieved.</i></p> |  |

Vetted By.....

Signature.....

Date.....

**BAIDOO BONSOE SENIOR HIGH TECHNICAL SCHOOL**  
**FORM TWO, SEMESTER ONE (1)**  
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| Learning Plan                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |                                                                                           |                 |                    |             |   |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------|--------------------|-------------|---|
| <b>Subject</b>                           | <i>Information &amp; Communications Technology</i>                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Week</b>       | 2                                                                                         | <b>Duration</b> | <i>120 minutes</i> | <b>Form</b> | 2 |
| <b>Week Ending</b>                       | 23 <sup>rd</sup> January, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                                                                                           |                 |                    |             |   |
| <b>Strand</b>                            | <i>ICTs in the Society</i>                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Sub-Strand</b> | <i>Organizing, Managing and Presenting Information Using Essential Productivity Tools</i> |                 |                    |             |   |
| <b>Content Standard</b>                  | <i>Demonstrate knowledge and understanding of analyzing numerical data using Spreadsheet software.</i>                                                                                                                                                                                                                                                                                                                                                                |                   |                                                                                           |                 |                    |             |   |
| <b>Learning Outcome(s)</b>               | <i>Utilize knowledge of Spreadsheet software to create functions for data analysis, interpret graphs and manipulate tables.</i>                                                                                                                                                                                                                                                                                                                                       |                   |                                                                                           |                 |                    |             |   |
| <b>Learning Indicator(s)</b>             | <ul style="list-style-type: none"> <li>• <i>Create and utilize formulae and basic functions (e.g., SUM, AVERAGE, MIN, MAX) for data analysis.</i></li> <li>• <i>Use advanced functions (e.g., IF, DATE, TIME).</i></li> </ul>                                                                                                                                                                                                                                         |                   |                                                                                           |                 |                    |             |   |
| <b>Essential Question(s)</b>             | <ul style="list-style-type: none"> <li>▪ <i>How do formulae and basic functions automate calculations responsibly and accurately?</i></li> <li>▪ <i>How can inclusive grouping and SEL support collaborative data analysis?</i></li> <li>▪ <i>What national values (accuracy, responsibility) are promoted through correct formula use?</i></li> <li>▪ <i>How do conditional and date/time functions support decision-making and responsible data use?</i></li> </ul> |                   |                                                                                           |                 |                    |             |   |
| <b>Pedagogical Strategies</b>            | <i>Talk for Learning</i><br><i>Experiential Learning</i><br><i>Brainstorming</i><br><i>Demonstration</i><br><i>Video-Assisted Learning</i><br><i>Group Work/Collaborative Learning</i><br><i>Think-Pair-Share, Questioning.</i><br><i>Practical Based Learning</i>                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |   |
| <b>Teaching &amp; Learning Resources</b> | <i>Laptops/Desktop computers</i><br><i>MS Excel/Google Sheets</i><br><i>Projector</i>                                                                                                                                                                                                                                                                                                                                                                                 |                   |                                                                                           |                 |                    |             |   |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | YouTube video<br>USB drive<br>Printed handouts                                                                                                       |
| <b>Key Notes on Differentiation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                      |
| <p><b>i. Learning Task:</b><br/>         Provide printed formulae steps and sample data for approaching proficiency learners.<br/>         Encourage high-proficiency learners to create their own datasets (e.g., family budget).<br/>         Discuss ethical data entry (accuracy, responsibility).<br/>         Printed IF syntax cards.</p> <p><b>ii. Pedagogical Exemplars:</b><br/>         Think-Pair-Share (mixed groups discuss formula uses).<br/>         Direct Instruction<br/>         Video-Assisted Learning (Excel functions tutorial).<br/>         Group practical (apply functions to marks sheet); SEL integration (peer feedback on calculations).</p> <p><b>iii. Key Assessments (DoK):</b><br/> <b>Level 1:</b><br/>         What symbol starts a formula?<br/>         What is the IF function used for?<br/> <b>Level 2:</b><br/>         Explain the SUM function and its purpose.<br/>         Explain IF syntax.<br/> <b>Level 3:</b><br/>         Use SUM and AVERAGE to calculate total and average marks in a dataset.<br/>         Use IF to assign pass/fail grades.<br/> <b>Level 4:</b><br/>         Create a formula for total/average unaided; evaluate why a formula might return an error (e.g., wrong cell reference).<br/>         Create nested IF for letter grades.</p> |                                                                                                                                                      |
| <b>Keywords</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Formula (=), SUM, AVERAGE, MIN, MAX, AutoFill, Relative Reference.</b><br><b>IF, Logical Test, Nested IF, DATE, TIME, Conditional Formatting.</b> |

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| <p align="center"><b>Lesson 1</b><br/> <b>Focal Areas:</b></p> <ol style="list-style-type: none"> <li>1. Basic formulae (=A1+B1) and SUM/AVERAGE functions; auto-fill.</li> <li>2. MIN/MAX functions; group practical on budget data.</li> </ol> |                         |
| <p align="center"><b>Main Lesson drawing on Concepts, Skills and Competencies to reinforce as in the Subject Teacher Manual</b></p>                                                                                                              |                         |
| <i>Teacher Activity</i>                                                                                                                                                                                                                          | <i>Learner Activity</i> |

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| <p><b><u>Starter Activity (10 minutes)</u></b></p> <p><b><i>Brainstorming:</i></b><br/> <i>Why is it better to use a formula instead of adding numbers manually?</i><br/> <i>Open Week 1 file.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b><u>Introductory Activity (15minutes)</u></b></p> <ul style="list-style-type: none"> <li>▪ <i>Discuss automation benefits (accuracy, speed); project formula bar.</i></li> <li>▪ <i>Demo MIN/MAX on marks data.</i></li> </ul> <p><b><u>Activity 1 (15 minutes)</u></b></p> <p><b><i>Demonstrate:</i></b><br/> <i>Simple formula (=A1+B1), then SUM (=SUM(B2:B10)).</i></p> <p><b><i>Guide practical:</i></b><br/> <i>Apply MIN/MAX to class marks.</i></p> <p><b><u>Activity 2 (15 minutes)</u></b></p> <ul style="list-style-type: none"> <li>▪ <i>Demonstration on basic functions and AutoFill.</i></li> <li>▪ <i>Group discussion: Create simple family budget sheet.</i></li> </ul> | <p><b><u>Introductory Activity (15minutes)</u></b></p> <ul style="list-style-type: none"> <li>▪ <i>Note = symbol and basic syntax.</i></li> <li>▪ <i>Enter data and apply functions.</i></li> </ul> <p><b><u>Activity 1</u></b></p> <ul style="list-style-type: none"> <li>▪ <i>Practice on provided marks dataset.</i></li> <li>▪ <i>Follow and replicate steps.</i></li> </ul> <p><b><u>Activity 2</u></b></p> <ul style="list-style-type: none"> <li>▪ <i>Watch and replicate steps.</i></li> <li>▪ <i>Collaborate and apply functions.</i></li> </ul> |
| <p><b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b><i>Level 1:</i></b><br/> <i>What symbol starts a formula?</i></p> <p><b><i>Level 2:</i></b><br/> <i>Explain the SUM function and its purpose.</i></p> <p><b><i>Level 3:</i></b><br/> <i>Use SUM and AVERAGE to calculate total and average marks in a dataset.</i></p> <p><b><i>Level 4:</i></b><br/> <i>Create a formula for total/average unaided; evaluate why a formula might return an error (e.g., wrong cell reference).</i></p>                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Lesson Closure</b></p> <p><b><i>In completing this part, refer to the Essential Questions to check that learning has taken place.</i></b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b><u>Activity (5 minutes)</u></b></p> <p><b><i>Recap:</i></b></p> <ul style="list-style-type: none"> <li>▪ <i>How does SUM promote responsibility in data handling</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| <ul style="list-style-type: none"> <li>How do functions support accurate and responsible analysis?</li> </ul>                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                   |
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| <p align="center"><b>Lesson 2</b></p>                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                   |
| <p align="center"><b>Focal Areas:</b></p> <ol style="list-style-type: none"> <li>IF function for pass/fail grading.</li> <li>Nested IF for letter grades; DATE/TIME functions.</li> </ol>                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                   |
| <i>Teacher Activity</i>                                                                                                                                                                                                                                                                                                                                                                                 | <i>Learner Activity</i>                                                                                                                                                                                                                                                           |
| <p><b>Starter <u>Activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Review lesson on functions.</li> <li>How can a computer decide pass/fail automatically?</li> </ul>                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                   |
| <p><b><u>Introductory activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Demo IF syntax<br/>(=IF(B2&gt;=50,"Pass","Fail")).</li> <li>Demo nested IF and DATE.</li> </ul> <p><b><u>Activity 1 (15 minutes)</u></b></p> <p><b>Guide practical:</b><br/>IF and Nested IF for grades (A/B/C/D/F).</p> <p><b><u>Activity 2 (15 minutes)</u></b><br/>Add DATE/TIME to budget sheet.</p> | <p><b><u>Introductory activity</u></b><br/>Learners must observe and note the syntax.</p> <p><b><u>Activity 1</u></b><br/>Students will apply IF and Nested IF to dataset.</p> <p><b><u>Activity 2 (15 minutes)</u></b><br/>Students will collaborate and present their work.</p> |
| <p align="center"><b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b></p>                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |
| <p><b>Level 1:</b><br/>What is the IF function used for?</p> <p><b>Level 2:</b><br/>Explain IF syntax.</p> <p><b>Level 3:</b><br/>Use IF to assign pass/fail grades.</p> <p><b>Level 4:</b><br/>Create nested IF for letter grades.</p>                                                                                                                                                                 |                                                                                                                                                                                                                                                                                   |
| <p align="center"><b>Lesson Closure</b></p> <p align="center"><i>In completing this part, refer to the Essential Questions to check that learning has taken place.</i></p>                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |

**Activity (10 minutes)**

***Recap:***

*Essential questions.*

**Reflection & Remarks**

***Reflection:***

- *Learners proud of using excel to that level*
- *Nested IF understood well*
- *There were some groups having syntax errors in Nested IF.*

***Remarks:***

*Lesson objectives are largely achieved.*

Vetted By.....

Signature.....

Date.....

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**FORM TWO, SEMESTER ONE (1)**  
**FACILITATOR NAME: BAAH EZEKIEL**

| Learning Plan                            |                                                                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |   |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------|--------------------|-------------|---|
| <b>Subject</b>                           | <i>Information &amp; Communications Technology</i>                                                                                                                                                                                                                                                                 | <b>Week</b>       | 3                                                                                         | <b>Duration</b> | <i>120 minutes</i> | <b>Form</b> | 2 |
| <b>Week Ending</b>                       | 30 <sup>th</sup> January, 2026                                                                                                                                                                                                                                                                                     |                   |                                                                                           |                 |                    |             |   |
| <b>Strand</b>                            | <i>ICTs in the Society</i>                                                                                                                                                                                                                                                                                         | <b>Sub-Strand</b> | <i>Organizing, Managing and Presenting Information Using Essential Productivity Tools</i> |                 |                    |             |   |
| <b>Content Standard</b>                  | <i>Demonstrate knowledge and understanding of analyzing numerical data using Spreadsheet software.</i>                                                                                                                                                                                                             |                   |                                                                                           |                 |                    |             |   |
| <b>Learning Outcome(s)</b>               | <i>Utilize knowledge of Spreadsheet software to create functions for data analysis, interpret graphs and manipulate tables.</i>                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |   |
| <b>Learning Indicator(s)</b>             | <i>Apply sorting and filtering operations to manipulate tables.</i>                                                                                                                                                                                                                                                |                   |                                                                                           |                 |                    |             |   |
| <b>Essential Question(s)</b>             | <ul style="list-style-type: none"> <li>▪ <i>How do sorting and filtering operations help organize and analyze large datasets responsibly and inclusively?</i></li> <li>▪ <i>How can GESI, SEL, and national values (e.g., responsibility, accuracy) be integrated when teaching spreadsheet basics?</i></li> </ul> |                   |                                                                                           |                 |                    |             |   |
| <b>Pedagogical Strategies</b>            | <i>Talk for Learning</i><br><i>Experiential Learning</i><br><i>Brainstorming</i><br><i>Demonstration</i><br><i>Questioning</i><br><i>Video-Assisted Learning</i><br><i>Group Work/Collaborative Learning</i><br><i>Think-Pair-Share, Questioning.</i>                                                              |                   |                                                                                           |                 |                    |             |   |
| <b>Teaching &amp; Learning Resources</b> | <i>Laptops/Desktop computers</i><br><i>MS Excel/Google Sheets</i><br><i>Projector</i><br><i>USB drive</i><br><i>Printed handouts/steps</i>                                                                                                                                                                         |                   |                                                                                           |                 |                    |             |   |
| <b>Key Notes on Differentiation</b>      |                                                                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |   |

**i. Learning Task:**

Group learners considering GESI, SEL, and cross-cutting issues (mixed-ability/mixed-gender groups). Provide printed sorting/filtering steps and sample data for approaching proficiency learners. High-proficiency learners create their own filtered datasets (e.g., school club membership). Discuss ethical filtering (avoiding bias).

**ii. Pedagogical Exemplars:**

Direct Instruction (demo Data tab); Practical-Based (manipulate sales table); Think-Pair-Share (discuss filter results); SEL (reflect on fair data representation).

**iii. Key Assessments (DoK):****Level 1:**

*What is the difference between sorting and filtering?*

**Level 2:**

*Explain how to sort data in ascending order.*

**Level 3:**

*Apply sorting and filtering to a dataset and explain the result.*

**Level 4:**

*Design a filtered table for decision-making (e.g., identify top-performing students) and justify choices ethically.*

|                 |                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------|
| <b>Keywords</b> | <b><i>Sort (ascending/descending), Filter, Data tab, Table format, Custom filter, Advanced filter.</i></b> |
|-----------------|------------------------------------------------------------------------------------------------------------|

| <b>Lesson 1</b><br><b>Focal Areas:</b><br><b>1. Sorting operations (ascending/descending) on single/multiple columns</b>                                                                                                                                                                                                                                                              |                                                                                                                                                                                          |
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| <b>Main Lesson drawing on Concepts, Skills and Competencies to reinforce as in the Subject Teacher Manual</b>                                                                                                                                                                                                                                                                         |                                                                                                                                                                                          |
| <b><i>Teacher Activity</i></b>                                                                                                                                                                                                                                                                                                                                                        | <b><i>Learner Activity</i></b>                                                                                                                                                           |
| <b><u>Starter Activity (10 minutes)</u></b><br><b><i>Brainstorming:</i></b><br><i>How does sorting help us find information faster in a long list (e.g., student marks)?</i>                                                                                                                                                                                                          |                                                                                                                                                                                          |
| <b><u>Introductory Activity (15minutes)</u></b> <ul style="list-style-type: none"> <li><i>Discuss importance of organizing large data;</i></li> <li><i>Project Data tab.</i></li> </ul><br><b><u>Activity 1 (15 minutes)</u></b><br><b><i>Demonstrate:</i></b> <ul style="list-style-type: none"> <li><i>Sorting single column (ascending marks) and multiple columns.</i></li> </ul> | <b><u>Introductory Activity (15minutes)</u></b><br>Observe demonstration and take note the sort button<br><br><br><b><u>Activity 1</u></b><br><i>Practice on provided marks dataset.</i> |

|                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u><b>Activity 2 (15 minutes)</b></u><br><i>Teacher demonstration on sorting techniques.</i>                                                                                                                                                                                                                                                                                                                               | <u><b>Activity 2</b></u><br><i>Watch and replicate steps.</i>                                                                                                                                                                                                     |
| <b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b>                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                   |
| <i>Level 1: What is sorting?</i><br><i>Level 2: Explain ascending vs descending sort.</i><br><i>Level 3: Apply sorting to a dataset.</i><br><i>Level 4: Sort multiple columns and explain why order matters.</i>                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                   |
| <p style="text-align: center;"><b>Lesson Closure</b></p> <p style="text-align: center;"><i><b>In completing this part, refer to the Essential Questions to check that learning has taken place.</b></i></p>                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                   |
| <u><b>Activity (5 minutes)</b></u><br><i>How does sorting promote accuracy and responsibility in data handling?</i>                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                   |
| <p style="text-align: center;"><b>Lesson 2</b></p> <p style="text-align: center;"><b>Focal Areas:</b><br/> <b>Filtering operations (basic/custom filters); group practical on sales/market data.</b></p>                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                   |
| <i><b>Teacher Activity</b></i>                                                                                                                                                                                                                                                                                                                                                                                             | <i><b>Learner Activity</b></i>                                                                                                                                                                                                                                    |
| <b>Starter <u>Activity (10 minutes)</u></b><br><ul style="list-style-type: none"> <li><i>Review functions on Dataset.</i></li> </ul>                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                   |
| <u><b>Introductory activity (10 minutes)</b></u><br><ul style="list-style-type: none"> <li><i>Demo Filter button and custom criteria.</i></li> </ul><br><u><b>Activity 1 (15 minutes)</b></u><br><br><b>Guide practical:</b><br><i>Apply basic and custom filters to student marks.</i><br><br><u><b>Activity 2 (15 minutes)</b></u><br><br><i>Filter market price data to find affordable items and discuss insights.</i> | <u><b>Introductory activity</b></u><br><i>Note how filters hide/show rows.</i><br><br><u><b>Activity 1</b></u><br><i>Follow steps and save work.</i><br><br><br><br><u><b>Activity 2 (15 minutes)</b></u><br><br><i>Collaborate and present filtered results.</i> |
| <b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b>                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                   |

**Level 4:**

- *Apply custom filter.*
- *Justify how filtering supports fair analysis (e.g., avoiding bias).*

**Lesson Closure**

***In completing this part, refer to the Essential Questions to check that learning has taken place.***

**Activity (10 minutes)**

***Recap:***

*How do filters help inclusive data analysis?*

**Reflection & Remarks**

***Reflection:***

- *Learners excited when filters revealed hidden insights (e.g., top students).*
- *Strong collaboration in groups.*
- *Some confusion with custom filters.*

***Remarks:***

*Good mastery of Learning Indicator 4.*

Vetted By.....

Signature.....

Date.....

**BAIDOO BONSOE SENIOR HIGH TECHNICAL SCHOOL**  
**FORM TWO, SEMESTER ONE (1)**  
**FACILITATOR NAME: BAAH EZEKIEL**

| Learning Plan                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------|-----------------|--------------------|-------------|----------|
| <b>Subject</b>                                                                                                                                                                                                   | <i>Information &amp; Communications Technology</i>                                                                                                                                                                                                                 | <b>Week</b>       | <i>4</i>                                                                                  | <b>Duration</b> | <i>120 minutes</i> | <b>Form</b> | <i>2</i> |
| <b>Week Ending</b>                                                                                                                                                                                               | 6 <sup>th</sup> February, 2026                                                                                                                                                                                                                                     |                   |                                                                                           |                 |                    |             |          |
| <b>Strand</b>                                                                                                                                                                                                    | <i>ICTs in the Society</i>                                                                                                                                                                                                                                         | <b>Sub-Strand</b> | <i>Organizing, Managing and Presenting Information Using Essential Productivity Tools</i> |                 |                    |             |          |
| <b>Content Standard</b>                                                                                                                                                                                          | <i>Demonstrate knowledge and understanding of analyzing numerical data using Spreadsheet software.</i>                                                                                                                                                             |                   |                                                                                           |                 |                    |             |          |
| <b>Learning Outcome(s)</b>                                                                                                                                                                                       | <i>Utilize knowledge of Spreadsheet software to create functions for data analysis, interpret graphs and manipulate tables.</i>                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |
| <b>Learning Indicator(s)</b>                                                                                                                                                                                     | <i>Generate and interpret graphs/charts (bar, pie, line) to visualize data.</i>                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |
| <b>Essential Question(s)</b>                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>▪ <i>How do charts help visualize data engagingly and responsibly?</i></li> <li>▪ <i>How can GESI, SEL, and national values (e.g., responsibility, accuracy) be integrated when teaching spreadsheet basics?</i></li> </ul> |                   |                                                                                           |                 |                    |             |          |
| <b>Pedagogical Strategies</b>                                                                                                                                                                                    | <i>Talk for Learning</i><br><i>Experiential Learning</i><br><i>Brainstorming</i><br><i>Demonstration</i><br><i>Questioning</i><br><i>Video-Assisted Learning</i><br><i>Group Work/Collaborative Learning</i><br><i>Think-Pair-Share, Questioning.</i>              |                   |                                                                                           |                 |                    |             |          |
| <b>Teaching &amp; Learning Resources</b>                                                                                                                                                                         | <i>Laptops/Desktop computers</i><br><i>MS Excel/Google Sheets</i><br><i>Projector, YouTube videos/OER (Excel interface tutorials)</i><br><i>USB drive</i><br><i>Printed datasets</i>                                                                               |                   |                                                                                           |                 |                    |             |          |
| <b>Key Notes on Differentiation</b>                                                                                                                                                                              |                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |
| <b><i>i. Learning Task:</i></b><br><i>Group learners considering GESI, SEL, and cross-cutting issues (mixed-ability/mixed-gender groups). Discuss honest interpretation. Learners create combination charts.</i> |                                                                                                                                                                                                                                                                    |                   |                                                                                           |                 |                    |             |          |

**ii. Pedagogical Exemplars:**

*Direct Instruction (demo Data tab); Practical-Based (manipulate sales table); Think-Pair-Share; SEL (reflect on fair data representation).*

**iii. Key Assessments (DoK):****Level 1:**

*Name three chart types.*

**Level 2:**

*Explain when to use a pie chart vs bar chart.*

**Level 3:**

*Generate a chart from data and label it correctly.*

**Level 4:**

*Create and interpret a chart; explain insights and limitations ethically.*

**Keywords**

***Chart (bar, pie, line), Recommended Charts, Axes, Titles, Legend, Data labels, Interpretation.***

**Lesson 1****Focal Areas:**

**Inserting and formatting basic charts (bar, pie).**

**Main Lesson drawing on Concepts, Skills and Competencies to reinforce as in the Subject Teacher Manual**

***Teacher Activity******Learner Activity*****Starter Activity (10 minutes)*****Brainstorming:***

*How do pictures help us understand numbers better?*

**Introductory Activity (15minutes)**

- *Discuss visualization benefits.*

**Activity 1 (15 minutes)*****Demonstrate:***

- *Recommended Charts (bar for marks comparison).*

**Activity 2 (15 minutes)**

*Teacher demonstration on chart formatting (titles, axes).*

**Introductory Activity (15minutes)**

*Observe chart types.*

**Activity 1**

*Practice inserting bar chart*

**Activity 2**

*Watch and replicate steps.*

**Assessment DoK aligned to the Curriculum and Subject Teacher Manual**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Level 1:</b> Name three chart types.</p> <p><b>Level 2:</b> Explain when to use bar vs pie chart.</p> <p><b>Level 3:</b> Generate a basic chart.</p> <p><b>Level 4:</b> Format and interpret a chart ethically.</p>                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                        |
| <p style="text-align: center;"><b>Lesson Closure</b></p> <p style="text-align: center;"><i>In completing this part, refer to the Essential Questions to check that learning has taken place.</i></p>                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                        |
| <p><b><u>Activity (5 minutes)</u></b></p> <p>How do charts promote honest data communication?</p>                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                        |
| <p style="text-align: center;"><b>Lesson 2</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                        |
| <p style="text-align: center;"><b>Focal Areas:</b></p> <p style="text-align: center;">Line charts, mini-project interpretation, group presentation.</p>                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                        |
| <b><i>Teacher Activity</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b><i>Learner Activity</i></b>                                                                                                                                                                                                         |
| <p><b>Starter <u>Activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Review filtering and sorting of data.</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                        |
| <p><b><u>Introductory activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Demo line chart for trends (e.g., prices over time).</li> </ul> <p><b><u>Activity 1 (15 minutes)</u></b></p> <p><b>Guide practical:</b><br/>Insert line chart on sales data.</p> <p><b><u>Activity 2 (15 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Mini-project: "Digital Devices Usage in Ghana" (entry, functions, charts).</li> <li>Interpret insights.</li> <li></li> </ul> | <p><b><u>Introductory activity</u></b><br/>Students must note trend analysis.</p> <p><b><u>Activity 1</u></b><br/>Apply and format..</p> <p><b><u>Activity 2 (15 minutes)</u></b><br/>Collaborate; present charts (SEL feedback)..</p> |
| <p style="text-align: center;"><b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b></p>                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                        |
| <p><b>Level 4:</b></p> <ul style="list-style-type: none"> <li>Generate/interpret chart; explain real-world insights and limitations.</li> </ul>                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                        |
| <p style="text-align: center;"><b>Lesson Closure</b></p> <p style="text-align: center;"><i>In completing this part, refer to the Essential Questions to check that learning has taken place.</i></p>                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                        |

**Activity (10 minutes)**

***Recap:***

*How do charts support responsible data presentation?*

**Reflection & Remarks**

***Reflection:***

- *Learners engaged in mini-project and proud of charts.*
- *Strong interpretation discussions.*
- *Some struggled with chart selection.*
- *More chart examples next term.*

***Remarks:***

- *Excellent completion of learning indicator 5.*
- *Portfolios show spreadsheet skills.*

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Signature.....

Date.....

**BAIDOO BONSOE SENIOR HIGH TECHNICAL SCHOOL**  
**FORM TWO, SEMESTER ONE (1)**  
**FACILITATOR NAME: BAAH EZEKIEL**

| Learning Plan                            |                                                                                                                                                                                                                                                                                                                                                          |                   |                                      |                 |                    |             |   |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------|-----------------|--------------------|-------------|---|
| <b>Subject</b>                           | <i>Information &amp; Communications Technology</i>                                                                                                                                                                                                                                                                                                       | <b>Week</b>       | 5                                    | <b>Duration</b> | <i>120 minutes</i> | <b>Form</b> | 2 |
| <b>Week Ending</b>                       | 13 <sup>th</sup> February, 2026                                                                                                                                                                                                                                                                                                                          |                   |                                      |                 |                    |             |   |
| <b>Strand</b>                            | <i>Digital Devices and Emerging Technologies</i>                                                                                                                                                                                                                                                                                                         | <b>Sub-Strand</b> | <i>Components of Digital Devices</i> |                 |                    |             |   |
| <b>Content Standard</b>                  | <i>Demonstrate knowledge and understanding of digital devices and their components.</i>                                                                                                                                                                                                                                                                  |                   |                                      |                 |                    |             |   |
| <b>Learning Outcome(s)</b>               | <i>Describe the main components of digital devices and their functions.</i>                                                                                                                                                                                                                                                                              |                   |                                      |                 |                    |             |   |
| <b>Learning Indicator(s)</b>             | <i>Identify and describe components of digital devices (e.g., CPU, RAM, storage, input/output devices).</i>                                                                                                                                                                                                                                              |                   |                                      |                 |                    |             |   |
| <b>Essential Question(s)</b>             | <ul style="list-style-type: none"> <li>▪ <i>What are the key components of digital devices?</i></li> <li>▪ <i>How do these components work together to enable computing?</i></li> <li>▪ <i>How can GESI, SEL, and national values (e.g., innovation, responsibility in device use) be integrated when teaching digital device components?</i></li> </ul> |                   |                                      |                 |                    |             |   |
| <b>Pedagogical Strategies</b>            | <i>Talk for Learning</i><br><i>Experiential Learning</i><br><i>Brainstorming</i><br><i>Demonstration</i><br><i>Questioning</i><br><i>Video-Assisted Learning</i><br><i>Group Work/Collaborative Learning</i><br><i>Think-Pair-Share</i><br><i>Questioning</i>                                                                                            |                   |                                      |                 |                    |             |   |
| <b>Teaching &amp; Learning Resources</b> | <i>Laptops/Desktop computers</i><br><i>External Hard Drive</i><br><i>Mouse</i><br><i>Projector</i><br><i>Smartphone</i>                                                                                                                                                                                                                                  |                   |                                      |                 |                    |             |   |
| <b>Key Notes on Differentiation</b>      |                                                                                                                                                                                                                                                                                                                                                          |                   |                                      |                 |                    |             |   |

**i. Learning Task:**

Group learners considering GESI, SEL, and cross-cutting issues (mixed-ability/mixed-gender groups). Start with brainstorming device uses in Ghana (e.g., mobile phones for farming apps). High-proficiency learners research additional components (e.g., GPU). Discuss responsible device maintenance.

**ii. Pedagogical Exemplars:**

Direct Instruction (demo Data tab); Group Practical (label components on diagram); Think-Pair-Share; SEL (teamwork).

**iii. Key Assessments (DoK):****Level 1:**

Name three components of a digital device.

**Level 2:**

Explain the role of the CPU.

**Level 3:**

Describe how RAM and storage differ in function.

**Level 4:**

Diagram a digital device and evaluate how components interact for a task (e.g., running an app).

**Keywords**

**Digital Device, CPU (Central Processing Unit), RAM (Random Access Memory), Storage (HDD/SSD), Input Devices (keyboard, mouse), Output Devices (monitor, printer).**

**Lesson 1****Focal Areas:**

**Main components of digital devices (CPU, RAM, storage).**

**Main Lesson drawing on Concepts, Skills and Competencies to reinforce as in the Subject Teacher Manual**

**Teacher Activity****Learner Activity****Starter Activity (10 minutes)****Brainstorming:**

*What devices do you use daily, and what makes them 'digital'?*

**Introductory Activity (15minutes)**

- Discuss digital vs analog.
- Project diagram of computer internals.

**Activity 1 (15 minutes)****Demonstrate:**

- CPU role (brain of device).

**Introductory Activity (15minutes)**

Observe and note basic components.

**Activity 1**

*Practice labeling on printed diagrams.*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>RAM (temporary memory).</li> </ul> <p><b><u>Activity 2 (15 minutes)</u></b><br/>Video demo on hardware teardown.</p>                                                                                                                                                                                                                                                                                                | <p><b><u>Activity 2</u></b><br/>Watch, pause/replay, identify components</p>                                                                                                                                     |
| <b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                  |
| <p><b>Level 1:</b> Name three components.<br/> <b>Level 2:</b> Explain CPU function.<br/> <b>Level 3:</b> Distinguish RAM vs storage.<br/> <b>Level 4:</b> Evaluate component interaction.</p>                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  |
| <p style="text-align: center;"><b>Lesson Closure</b><br/> <i>In completing this part, refer to the Essential Questions to check that learning has taken place.</i></p>                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                  |
| <p><b><u>Activity (5 minutes)</u></b><br/>How do components enable responsible device use?</p>                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |
| <b>Lesson 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                  |
| <b>Focal Areas:</b><br>Input/output devices; group practical on device assembly.                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                  |
| <b><i>Teacher Activity</i></b>                                                                                                                                                                                                                                                                                                                                                                                                                             | <b><i>Learner Activity</i></b>                                                                                                                                                                                   |
| <p><b>Starter <u>Activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Quick quiz on Lesson 1 components.</li> </ul>                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                  |
| <p><b><u>Introductory activity (10 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Demo input (keyboard) and output (monitor).</li> </ul> <p><b><u>Activity 1 (15 minutes)</u></b></p> <p><b>Guide practical:</b><br/>Label input/output on real devices or diagrams.</p> <p><b><u>Activity 2 (15 minutes)</u></b></p> <ul style="list-style-type: none"> <li>Diagram a full device.</li> <li>Present how components work together.</li> </ul> | <p><b><u>Introductory activity</u></b><br/>Note functions.</p> <p><b><u>Activity 1</u></b><br/>Discuss uses.</p> <p><b><u>Activity 2 (15 minutes)</u></b><br/>Collaborate and share insights (SEL feedback).</p> |
| <b>Assessment DoK aligned to the Curriculum and Subject Teacher Manual</b>                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                  |

**Level 4:**

- *Create device diagram.*
- *Explain integration for Ghanaian context (e.g., mobile banking).*

**Lesson Closure**

*In completing this part, refer to the Essential Questions to check that learning has taken place.*

**Activity (10 minutes)*****Recap:***

*How does understanding components promote innovation?*

**Reflection & Remarks*****Reflection:***

- *Hands-on labeling engaged learners.*
- *Video teardown clarified concepts.*
- *Limited hardware samples.*
- *Visual aids helped.*
- *Bring more samples next week.*

***Remarks:***

*Solid progress on Learning Indicator.*

Vetted By.....

Signature.....

Date.....